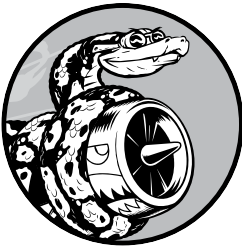


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GENERATING DATA



Data visualization is the use of visual representations to explore and present patterns in datasets. It's closely associated with *data analysis*, which uses code to explore the patterns and connections in a dataset. A dataset can be a small list of numbers that fits in a single line of code, or it can be terabytes of data that include many different kinds of information.

Creating effective data visualizations is about more than just making information look nice. When a representation of a dataset is simple and visually appealing, its meaning becomes clear to viewers. People will see patterns and significance in your datasets that they never knew existed.

Fortunately, you don't need a supercomputer to visualize complex data. Python is so efficient that with just a laptop, you can quickly explore datasets containing millions of individual data points. These data points don't